

Xiaoxuan(Andrina) Zhang

Languages: English(Fluent), Chinese (Fluent) and French (Intermediate)
Eligible to work for any U.S. employer | Willing to relocate and travel
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Portfolio: <https://andrinazxx.github.io/portfolio.html>

EDUCATION

Massachusetts Institute of Technology

September 2026 - May 2028

M.S. in Media Arts & Sciences Focus: Interactive software, immersive experience **Award:** Full Graduate Fellowship

University of California – San Diego

September 2020 - June 2024

B.S. Cognitive Science specialized in Machine Learning & Neural Computation

Overall GPA: 3.84

B.A. Interdisciplinary Computing and the Arts - Computer Music & Music Technology

Major GPA: 4.0

Minor in Computer Science & Engineering

WORK EXPERIENCE

Software Engineer Intern – Digital Signal Processing – Bose Professional, Hopkinton, MA May 2026 – Present

- Co-architected modular DSP pipelines with senior engineers; prototyped real-time algorithms (tonal volume control, pilot tone detection) in Python and refactored into production-grade C++ for stadium-scale systems.
- Analyzed acoustic data via custom Python helper scripts to ensure real-time signal integrity, leveraging LLM-assisted tools (GitHub Copilot) to accelerate C++ development and test case generation.

Research Assistant – Massachusetts Institute of Technology - Media Lab

October 2025 – Present

- Designing and building a multimodal AI interactive pattern generation system for a top luxury fashion brand, featuring ferrofluid and electromagnetic reactivity controlled by an audio-AI system.
- Created Max/MSP software systems and interface with Flask API / Python for Dallas Symphony Orchestra.

Software Engineer – Product Innovation – Bose Corporation, Framingham, MA

July 2024 – September 2025

- Performed data wrangling and analysis on audio machine learning models for key metrics optimization.
- Developed a Python-based prototype and automation framework to interface with Embedded Linux devices ([soundbars/speakers](#)) via SSH/Serial, to collect acoustic data.
- Authored comprehensive SOPs and data collection protocols for third-party testing partners.
- Gained firmware experience with STM32 (C / RTOS); practiced signal validation via oscilloscope testing.
- Actively coordinated with cross-functional teams using Jira/Confluence within an Agile/Scrum environment.

Software Research Engineer Intern – Qualcomm Institute, San Diego, CA

July 2023 – June 2024

- Developed a real-time beamforming system in Pure Data with a custom C++ plugin, achieving 11ms latency; scaled and validated on a 62-speaker, 4m wavefield synthesis setup.
- Integrated Kinect V2 depth sensing for real-time, user-tracked beam steering, modularizing the sensor pipeline for interactive installations.
- Built Python web scrapers to collect HRTF datasets; performed MATLAB-based spatial audio analysis.

Robotic Product Manager – Machani Robotics, San Diego, CA

January 2024 – April 2024

- Spearheaded and designed the product launch of an embodied AI robot to improve accessibility to quality mental health aid among students in a chain of universities.

SELECTED PROJECTS

Synthesizer AU/VST3 plug-in with JUCE and C++ for any DAWs [\[Source Code\]](#)

- Built a custom JUCE wavetable synthesizer, implementing **0–10 fold wavefolding**, **ADSR envelopes**, and a **nonlinear exponent shaper** to enable wide timbral variation and expressive modulation with visualizer.

Music Genre Classification with kNN, SVM, CNN and RNN implementation [\[Technical Report\]](#)

- Applied Exploratory Data Analysis and Data Visualization, after collected dataset and wrangled the data.
- Implemented **supervised** and **unsupervised** learning techniques and **deep learning** algorithms – Convolutional Neural Network and Recurrent Neural Network in Python (PyTorch, scikit learn, seaborn...).
- Designed the models and tested the algorithm and fine-tuned the weights and hyperparameters on GPU.

SKILL SET

Technical

Python, C++, Java, PureData, Max/MSP, MATLAB, C, JUCE, Xcode, LaTeX, Version Control / SQL, Git, XML, Digital Signal Processing, EEG Lab, RaspberryPi, OpenCV, CAD, Laser Cut

Creative

Ableton Live, Audacity, Reaper, Pro Tools, Final Cut, Adobe Photoshop / InDesign, Canva